

EST.
2019
PUNE, INDIA



CyTOS

CYCLE TIME OPTIMISING SOLUTIONS

COMPANY PROFILE

ENGINEERING SOLUTIONS FOR MANUFACTURING

Engineering advanced CNC machines,
automation solutions and industrial systems
for a more productive world.



CNC
TECHNOLOGY



INDUSTRIAL
AUTOMATION



ROBOTICS



FIXTURES &
TOOLING



CONTROL
SYSTEMS



EDUCATIONAL
SOLUTIONS



**BUILT TO WORK.
ENGINEERED TO LAST.**



RELIABLE



ROBUST



PRODUCTIVE



SUPPORTIVE



www.cytos.co.in

ENGINEERING SOLUTIONS THAT DRIVE PROGRESS

CyTOS is an engineering company that designs, develops and delivers advanced CNC machines, automation solutions, industrial systems, fixtures and control panels for a wide range of industries.

Since our inception in 2019, our focus has been simple – deliver reliable, robust and high performance solutions that help our customers manufacture with ease and productivity.



15+

DEDICATED
PROFESSIONALS



3000

SQ. FT.
PRODUCTION FACILITY



WHAT WE DO



CNC MACHINE SOLUTIONS

High performance CNC machines for diverse applications.



FIXTURE DESIGN & MANUFACTURING

Custom fixtures and tooling for precision and productivity.



INDUSTRIAL AUTOMATION

Automation solutions that improve efficiency and reduce cycle time.



ROBOTICS SOLUTIONS

Robotic systems for welding, assembly, handling and special applications.



CUSTOM INDUSTRIAL CONTROL PANELS

Design and manufacturing of control panels for industrial applications.



EDUCATIONAL CNC SOLUTIONS

Training and educational CNC equipment for future engineers.



CNC CONTROLLER & SOFTWARE

In-house CNC controller hardware and control software development.



SERVICE & SUPPORT

Installation, training and after-sales support for long term performance.

OUR ENGINEERING STRENGTH



ENGINEERING DRIVEN DESIGN

Every machine is designed with complete engineering knowledge, calculations and simulation analysis.



SAFETY FIRST

We design with safety factor of 2 to ensure strength, stability and long life.



BUILT FOR 24X7 OPERATION

Our machines are heavier by design but built to work without stoppage 24X7.



RIGOROUS TESTING

Every machine goes through heavy trials and tests. We correct even an aesthetically simple flaw.



COMPLETE SOLUTIONS

From concept to commissioning, we deliver complete turnkey solutions with support.



FOCUS

Customer centric solutions that create real value.



INNOVATION

Continuous improvement through technology and creativity.



QUALITY

Commitment to high quality in every part and every process.



TEAMWORK

A skilled team working together for excellence.



CYCLE TIME OPTIMISING SOLUTIONS | CyTOS

Engineering Solutions That Drive Progress

CYCLE TIME OPTIMISING SOLUTIONS (CyTOS) is an engineering and manufacturing solutions company established in **December 2019**, based in **Pune, Maharashtra, India**.

We specialise in the design and development of **CNC machines, PCB drilling and routing machines, CNC engraving and routing solutions, fixtures, industrial automation systems, educational CNC machines, and customised industrial control panels**.

Our focus is to develop reliable, robust, and practical engineering solutions that help manufacturers improve productivity, reduce production cycle time, minimise labour dependency, and achieve consistent performance.

With a dedicated team of approximately **15 professionals** and a **3,000 sq. ft. production facility**, CyTOS combines engineering expertise, practical manufacturing knowledge, and in-house CNC controller and software development capabilities to deliver complete solutions—from concept and design to manufacturing, commissioning, installation, training, and after-sales support.



OUR STORY

Why CyTOS Was Started ?

CyTOS was established after identifying a recurring challenge faced by manufacturers across several industries.

In the PCB manufacturing sector, CNC machines are critical production equipment, and unexpected machine breakdowns can significantly affect production. At the time, many CNC machines available in the Indian market were imported, particularly from China. In several cases, machine controllers and software interfaces were difficult to operate and service locally, creating challenges in troubleshooting, maintenance, and breakdown support.

We also observed similar challenges in **wood, aluminium, PCB, and other soft-material routing industries**, where manufacturers required reliable machines that were easy to operate, maintain, and service.

The available alternatives were often either expensive or did not provide the desired combination of quality, reliability, serviceability, and performance.

This led to the creation of CyTOS.

We began developing and manufacturing our own CNC machines from the ground up, with a clear objective:

To build reliable, robust, high-performance machines that are practical to operate, easy to maintain, and capable of supporting demanding industrial production.

Today, CyTOS has expanded beyond CNC machines into **fixture design and manufacturing, industrial automation, educational CNC solutions, and customised industrial control panels**, while continuing to focus on solving real manufacturing challenges through engineering.

OUR VISION

Building a Globally Capable Manufacturing Ecosystem

Our vision is to establish a **global presence in industrial and educational sectors** by developing innovative, reliable, and accessible engineering solutions.

We aim to contribute to the growth of manufacturing industries by enabling businesses and educational institutions to adopt advanced technologies with confidence.

OUR MISSION

Enabling Manufacturing with Ease and Productivity

Our mission is to contribute to the development of the global manufacturing industry by creating solutions that enable manufacturers to **produce with greater ease, higher productivity, improved reliability, and reduced cycle time**.

We continuously work toward developing engineering solutions that address practical industrial challenges while delivering long-term value to our customers.

OUR CORE VALUES

Engineering with Purpose

Our core value is simple:

To provide the best possible engineering solution for every industrial manufacturing challenge.

We believe that every manufacturing problem deserves a practical, reliable, and well-engineered solution.

Our approach is built around:

- Customer-Centric Thinking
- Engineering Excellence
- Reliability
- Innovation
- Quality
- Continuous Improvement
- Long-Term Customer Support.

WHAT WE DO

Our Engineering & Manufacturing Solutions

1. CNC MACHINE SOLUTIONS

We design and manufacture CNC machines for a wide range of applications, including:

- PCB drilling
- PCB routing
- CNC engraving
- CNC routing
- Soft-material machining
- Aluminium machining
- Industrial PCB production
- CNC applications for educational institutions

Our machines are designed for reliable and continuous operation, with solutions available for both **prototyping** and **full-scale industrial production**.

We also offer customised CNC machine designs based on specific workpiece sizes, production requirements, spindle configurations, automation levels, and application requirements.

2. PCB CNC DRILLING & ROUTING MACHINES

CyTOS develops specialised CNC machines for PCB drilling and routing applications.

Our machines are designed to support applications ranging from **PCB prototyping** to **full-scale industrial PCB production**.

Depending on the model and configuration, our machines can be equipped with:

- High-speed spindles
- Multiple spindle configurations
- Automatic Tool Changer (ATC)
- Large working areas
- High-speed machine movement
- Customised machine configurations

Our machines can achieve spindle speeds of up to **60,000 RPM**, with drilling capabilities down to approximately **0.2 mm drill diameter** on suitable materials and applications.

Selected machines can be configured with vacuum tables and vacuum dust extraction systems as per customer requirements.



3. FIXTURE DESIGN & MANUFACTURING

CyTOS provides complete fixture engineering solutions for industrial manufacturing applications.

We work with:

- Welding fixtures
- Assembly fixtures
- Inspection fixtures
- Machining fixtures
- Production fixtures
- Robotic fixtures
- Special-purpose fixtures

Our fixtures are designed according to the specific application, production process, and required accuracy.



We commonly work with **mild steel and stainless steel**, while material selection is determined according to the application, design requirements, operating conditions, and expected service life.

Our fixture development process can begin from:

- 3D CAD models
- Physical samples
- Customer drawings
- Existing components
- Application requirements

Our services include complete fixture design, fabrication, assembly, and finishing. Machining operations, when required, are managed through specialised external machining partners.

Example Application

One of our fixture solutions was developed for a welding application involving an **L-shaped component** that required welding on both horizontal and vertical surfaces.

To improve the welding process, we designed a fixture that could rotate the component orientation by **90 degrees**, allowing the robot to perform welding on both perpendicular surfaces while maintaining a convenient horizontal working position.

Such solutions help improve:

- Machining and assembly accuracy
- Production consistency
- Operator convenience
- Production cycle time

We have worked on multiple fixture requirements for **fabrication companies, CNC machining companies, PCB manufacturers, and other industrial customers.**



4. INDUSTRIAL AUTOMATION

CyTOS designs and builds complete industrial automation systems tailored to specific manufacturing requirements.

Our automation capabilities include:

- Material handling
- Pick-and-place systems
- Robotic automation
- Conveyor systems
- Assembly automation
- Process automation
- Inspection automation
- Special Purpose Machines (SPMs)

We provide complete solutions covering:



Concept → Engineering → Design → Control System → Fabrication → Assembly → Programming → Commissioning

We commonly work with automation platforms and components from leading manufacturers such as **Delta and Mitsubishi**, while also supporting other PLC and control platforms according to customer requirements.

Our automation solutions are developed to:

- Reduce labour dependency
- Increase production capacity
- Improve process consistency
- Reduce cycle time
- Improve workplace safety
- Increase Productivity

Featured Automation Application

One of our notable automation projects involved a **robotic adhesive dispensing and component placement system for automotive roof applications**.

The system used a robot to dispense hot adhesive while pneumatic actuators and grippers positioned and placed corresponding components at defined locations on the vehicle roof.

This type of integrated automation solution demonstrates our capability to combine **robotics, pneumatic systems, process control, and customised machine engineering** into a complete production solution.



5. EDUCATIONAL CNC SOLUTIONS

CyTOS develops CNC machines specifically for educational institutions and technical training environments.

Our objective is to provide students with practical, hands-on exposure to CNC technology before they enter the manufacturing industry.

OUR EDUCATIONAL CNC PLATFORM INCLUDES:

-  CNC Machines
-  CNC Controller
-  CAM Software
-  Installation
-  Teacher Training
-  Warranty
-  After-sales Support



OUR MACHINES ARE SUITABLE FOR WORKING WITH MATERIALS SUCH AS:



 EMPOWERING THE NEXT GENERATION OF ENGINEERS AND INNOVATORS

PCBE3020

Our PCBE3020 educational CNC machine features a **300 x 200 mm** working area and is available with **Automatic Tool Changer (ATC)** capability.

This is a compact yet powerful educational CNC platform, ideal for institutions to focus on designing and practical applications with contemporary machine downtime.



PRACTICAL
LEARNING



INDUSTRY
RELEVANT SKILLS



INNOVATE
& CREATE



BUILD A STRONG
FOUNDATION

6. CUSTOMISED INDUSTRIAL CONTROL PANELS

CyTOS designs and builds customised industrial control panels for a wide range of industrial applications.

Our control panel solutions are developed according to individual machine and system requirements.

We work with:

- PLC-based control systems
- Industrial automation systems
- Machine control systems
- Custom electrical control architectures
- Multiple PLC platforms

Our control panels are designed with attention to engineering standards and can be prepared according to applicable **IEC and CE requirements**, depending on project specifications and customer requirements.

While the sheet metal required for panel construction is outsourced, **panel design, assembly, wiring, integration, and testing** are managed by our team.



OUR CNC PRODUCT PORTFOLIO

Precision. Performance. Productivity.

PCB30

-  **Working Area**
300 × 300 mm
-  **Spindle Configuration**
1 Spindle
-  **Standard Spindle Speed**
40,000 RPM
-  **ATC**
Available
-  **Primary Application**
PCB Prototyping

The PCB30 is a compact CNC solution designed for PCB prototyping and precision applications where a smaller working area is required.



PCB60

-  **Working Area**
600 × 600 mm
-  **Spindle Configuration**
1 or 2 Spindles
-  **Standard Spindle Speed**
40,000 RPM
-  **ATC**
Available
-  **Primary Applications**
PCB Drilling, PCB Routing and Soft-Material Applications

The PCB60 is designed for applications requiring a larger working area and increased production flexibility.



PCB12

-  **Working Area**
1,200 × 1,200 mm
-  **Spindle Configuration**
3 Spindles
-  **Standard Spindle Speed**
60,000 RPM
-  **ATC**
Available
-  **Primary Applications**
PCB Drilling, PCB Routing, Aluminium and Other Suitable Materials

The PCB12 is a large-format CNC platform designed for high-productivity applications.

The standard platform allows the spindle to operate across the 1,200 × 1,200 mm working area with individual spindle working configurations and can also be provided as part of customized machine development.



HIGH PRECISION
Superior accuracy for perfect results



HIGH SPEED
Up to 60,000 RPM spindle speed



ATC AVAILABLE
Automatic Tool Change for uninterrupted operation



RELIABLE & ROBUST
Built for continuous 24x7 industrial operation



VERSATILE APPLICATIONS
Ideal for PCB, aluminium and soft materials



ROUTER SERIES

HIGH PERFORMANCE CNC ROUTING SOLUTIONS

ROUTER 4 × 4



Working Area
4 × 4 ft



Spindle Speed
24,000 RPM



ATC
Available

SUITABLE MATERIALS



PCB



MDF



Wood



Acrylic



Nylon



Brass



Copper



Aluminium

ROUTER 8 × 8



Working Area
8 × 8 ft



Spindle Speed
24,000 RPM



ATC
Available

SUITABLE MATERIALS



PCB



MDF



Wood



Acrylic



Nylon



Brass



Copper



Aluminium



HIGH PRECISION
Superior accuracy
for perfect results



HIGH SPEED
24,000 RPM spindle for
maximum productivity



**HEAVY DUTY
CONSTRUCTION**
Built for long life and
continuous performance



24 × 7 OPERATION
Designed for non-stop
industrial applications



EASY MAINTENANCE
User friendly design for
effortless maintenance

CUSTOM CNC MACHINE DEVELOPMENT **Built Around Your Requirements**

We understand that every production environment is different.

Therefore, CyTOS offers customised CNC machine development based on:

- Working area
- Material requirements
- Spindle configuration
- Number of spindles
- ATC requirements
- Production volume
- Machine speed
- Application-specific tooling
- Automation requirements

We have successfully developed customised CNC machines, including a **1,200 × 1,200 mm, three-spindle CNC machine with ATC on each spindle**, capable of achieving travel speeds of up to **15,000 mm/min**.

Our engineering team works closely with customers to understand their application and develop a machine suited to their specific production requirements.

OUR ENGINEERING STRENGTH **Engineering-Driven Design**

At CyTOS, engineering is at the core of everything we build.

We design our machines and systems using comprehensive engineering knowledge, supported by appropriate calculations, design validation, and simulation analysis.

Every component is designed with consideration for:

- Structural strength
- Operating loads
- Machine stability
- Safety factors
- Service life
- Maintainability
- Manufacturing practicality

Our engineering approach ensures that the final machine is not only functional but also designed for dependable long-term operation.



OUR MANUFACTURING CAPABILITIES

CyTOS has a dedicated **3,000 sq. ft. production facility** supported by a team of approximately **15 professionals**.

Our capabilities include:

In-House Capabilities

- Machine frame and structural fabrication
- Mechanical assembly and Machine wiring
- CNC controller development
- CNC control software development
- Machine wiring
- Machine commissioning
- Final testing
- Installation
- Customer training

Supporting External Capabilities

Specialized processes such as CNC machining of certain machine components and sheet-metal fabrication for control panels are managed through experienced external manufacturing partners.

This combination allows us to maintain strong control over **engineering, assembly, integration, testing, and final machine performance**, while leveraging specialised manufacturing capabilities where required.

DESIGN & ENGINEERING INFRASTRUCTURE

Our dedicated design department is equipped with powerful computer systems for:

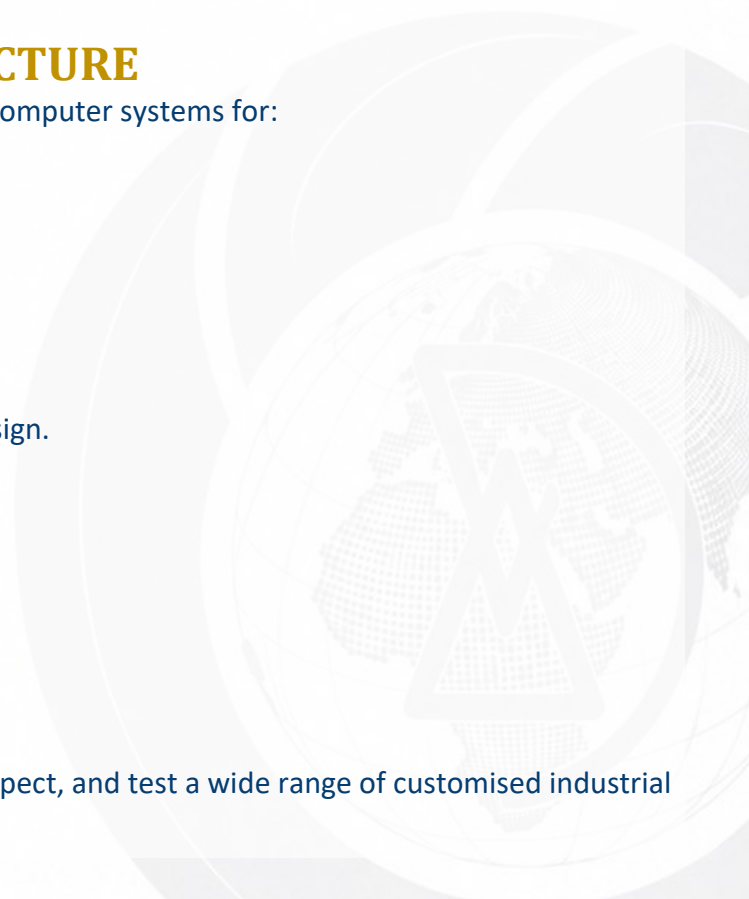
- 3D machine design
- Mechanical design
- Engineering calculations
- Machine analysis
- Simulation
- Product development

We use **SolidWorks** for 3D CAD modelling and engineering design.

Our production infrastructure also includes:

- Welding setup
- Measurement and inspection tools
- Vernier callipers
- Gauges
- Complete mechanical assembly toolkit

This infrastructure enables our team to develop, assemble, inspect, and test a wide range of customised industrial solutions.



OUR INDUSTRIES

CyTOS has experience working with customers and applications across multiple industrial sectors, including:

- PCB Manufacturing
- Packaging Industry
- Textile Supplier Industry
- Wood Industry
- Construction Material Industry
- Fabrication Industry
- CNC Machining Industry
- Educational Institutions
- Automotive-related Applications
- Industrial Manufacturing

Our cross-industry experience enables us to understand different manufacturing environments and develop solutions according to specific application requirements.

WHY CHOOSE CYTOS?

Reliable. Robust. Engineered for Performance.

1. ENGINEERING-DRIVEN SOLUTIONS

We design machines and systems based on engineering calculations, analysis, and practical manufacturing knowledge.

2. COMPLETE SOLUTION APPROACH

From concept development to design, manufacturing, assembly, commissioning, installation, and training, we provide integrated solutions.

3. CUSTOMISATION

We develop machines, fixtures, automation systems, and control solutions according to specific customer requirements.

4. RELIABILITY

Our machines are designed with a focus on dependable performance and long-term operation.

5. ROBUST CONSTRUCTION

We prioritise structural strength, stability, and durability in our machine designs.

6. IN-HOUSE CNC TECHNOLOGY

Our ability to develop CNC controllers and CNC control software gives us greater control over machine functionality and allows us to build solutions tailored to customer requirements.

7. COMPLETE CUSTOMER SUPPORT

We provide installation, operator training, commissioning, warranty support, after-sales service, and annual maintenance/service contracts.

OUR APPROACH FROM CHALLENGE TO SOLUTION

We follow a structured engineering approach:

01 — Understand

We study the customer's manufacturing challenge and production requirements.

02 — Analyse

We evaluate the application, materials, process, productivity requirements, and technical constraints.

03 — Design

We develop engineering concepts and detailed 3D CAD designs.

04 — Engineer

We perform calculations, design validation, and simulation analysis where required.

05 — Build

We manufacture, fabricate, assemble, wire, and integrate the solution.

06 — Test

We conduct detailed trials and functional testing.

07 — Commission

We install and commission the machine or system at the customer's facility.

08 — Train & Support

We train operators and provide ongoing service and technical support.

AFTER-SALES SUPPORT

Supporting Your Machine Beyond Delivery

Our relationship with customers does not end after machine delivery.

We provide:

- Machine installation and Operator training
- Commissioning
- Teacher training for educational solutions
- Warranty support
- Annual maintenance/service contracts
- Technical assistance
- After-sales support

Our objective is to ensure that customers can operate their machines confidently and achieve long-term productivity.

OUR COMMITMENT

At CyTOS, we believe that a successful engineering solution is one that delivers value throughout its entire lifecycle.

Our commitment is to develop solutions that are:

Reliable
Robust`
Practical
Maintainable
Productive
Engineered for Long-Term Performance

We continuously strive to understand our customers' challenges and transform those challenges into practical engineering solutions.

LOOKING AHEAD

Engineering the Future of Manufacturing

CyTOS is continuously expanding its capabilities across **CNC technology, industrial automation, robotics, fixture engineering, educational solutions, and industrial control systems.**

Our long-term goal is to establish CyTOS as a trusted engineering partner for manufacturing companies and educational institutions in India and across the world.

We envision a future where manufacturers can access reliable engineering technology that enables them to **produce better, faster, and with greater confidence.**

OUR PROMISE

we don't just build machines. We engineer solutions to make manufacturing easier, more productive, and more reliable.

CONTACT US

CYCLE TIME OPTIMISING SOLUTIONS

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CNC MACHINES | FIXTURES | INDUSTRIAL AUTOMATION | EDUCATIONAL CNC | CONTROL PANELS
Engineering Solutions That Drive Progress



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ENGINEERING SOLUTIONS
THAT DRIVE **PRODUCTIVITY.**
PARTNERSHIPS THAT **LAST.**



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